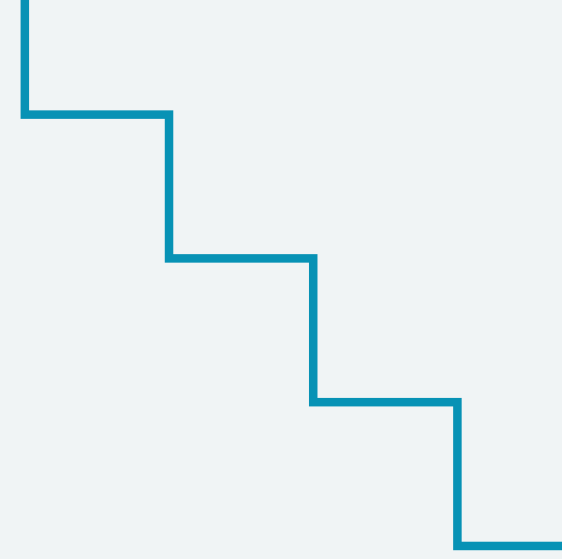
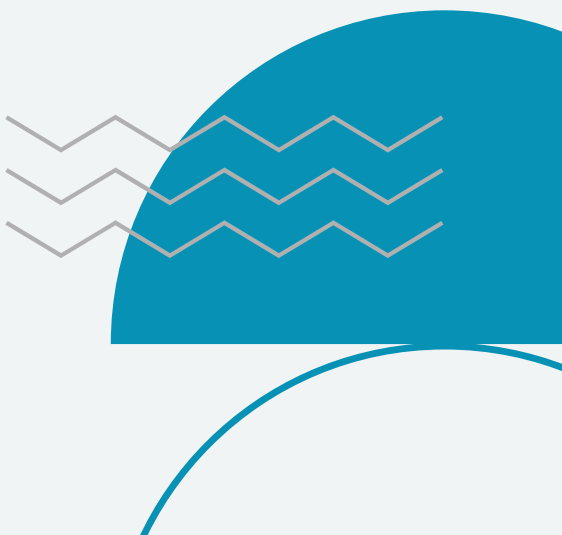




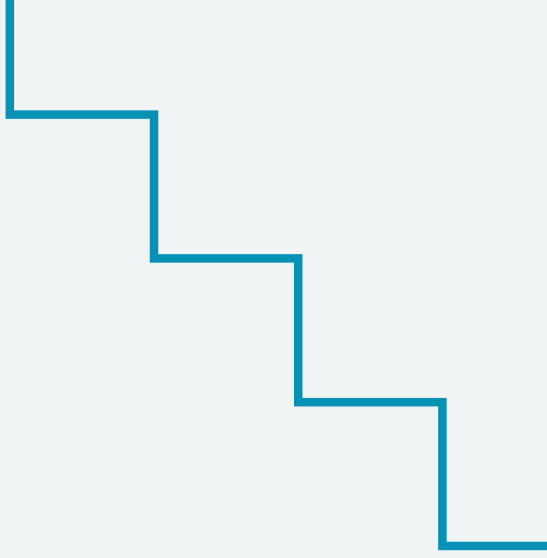
ROMI

Robotic Medical Intelligence





THE PROBLEM

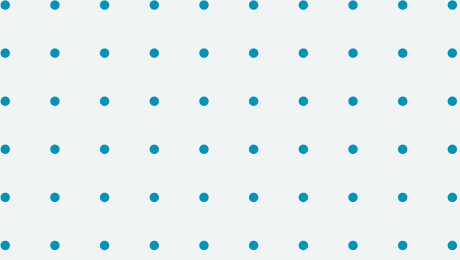


1. Problem Statement:

- Manual reception processes in Egyptian clinics and hospitals cause severe overcrowding, long waiting times, and administrative burnout, leading to a poor overall patient experience.
- 

2. Target Users

- Primary Users: Clinic Owners & Healthcare Administrators (seeking cost reduction and efficiency).
 - Secondary Users: Patients (seeking fast, secure, and hassle-free check-ins).
 - End Users: Medical Receptionists (needing relief from repetitive manual tasks).
- 



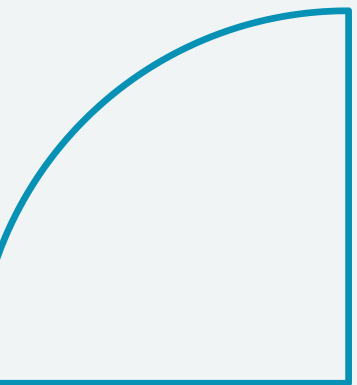
RESEARCH & VALIDATION

1. Research Findings

- Overcrowding Crisis: Local healthcare studies (e.g., in major facilities like Ain Shams University Hospitals) indicate that manual check-ins and administrative bottlenecks contribute to a 46% increase in patient waiting times.
- Wasted Resources: Medical receptionists spend over 60% of their shifts on routine verification rather than actual patient care.

2. References

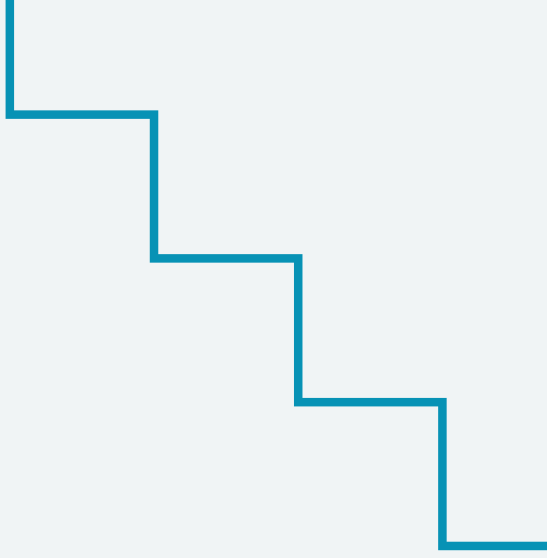
- Published healthcare reports on patient flow in Egyptian hospitals (Ain Shams University studies).
- Direct interviews and operational observations from local private clinics.





PROPOSED SOLUTION

(ROMI)

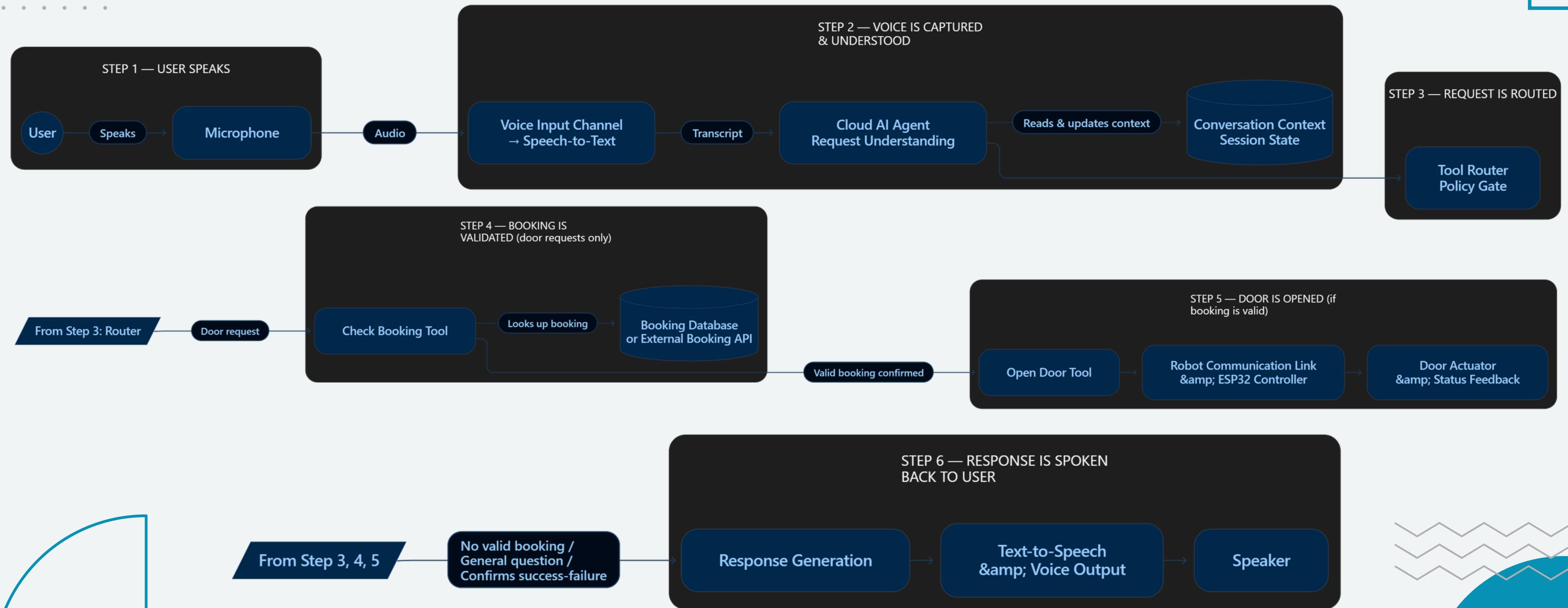


1. What is ROMI? An autonomous, voice-powered AI receptionist that eliminates clinic queues by validating appointments via natural conversation and unlocking doors via IoT, with zero human intervention.

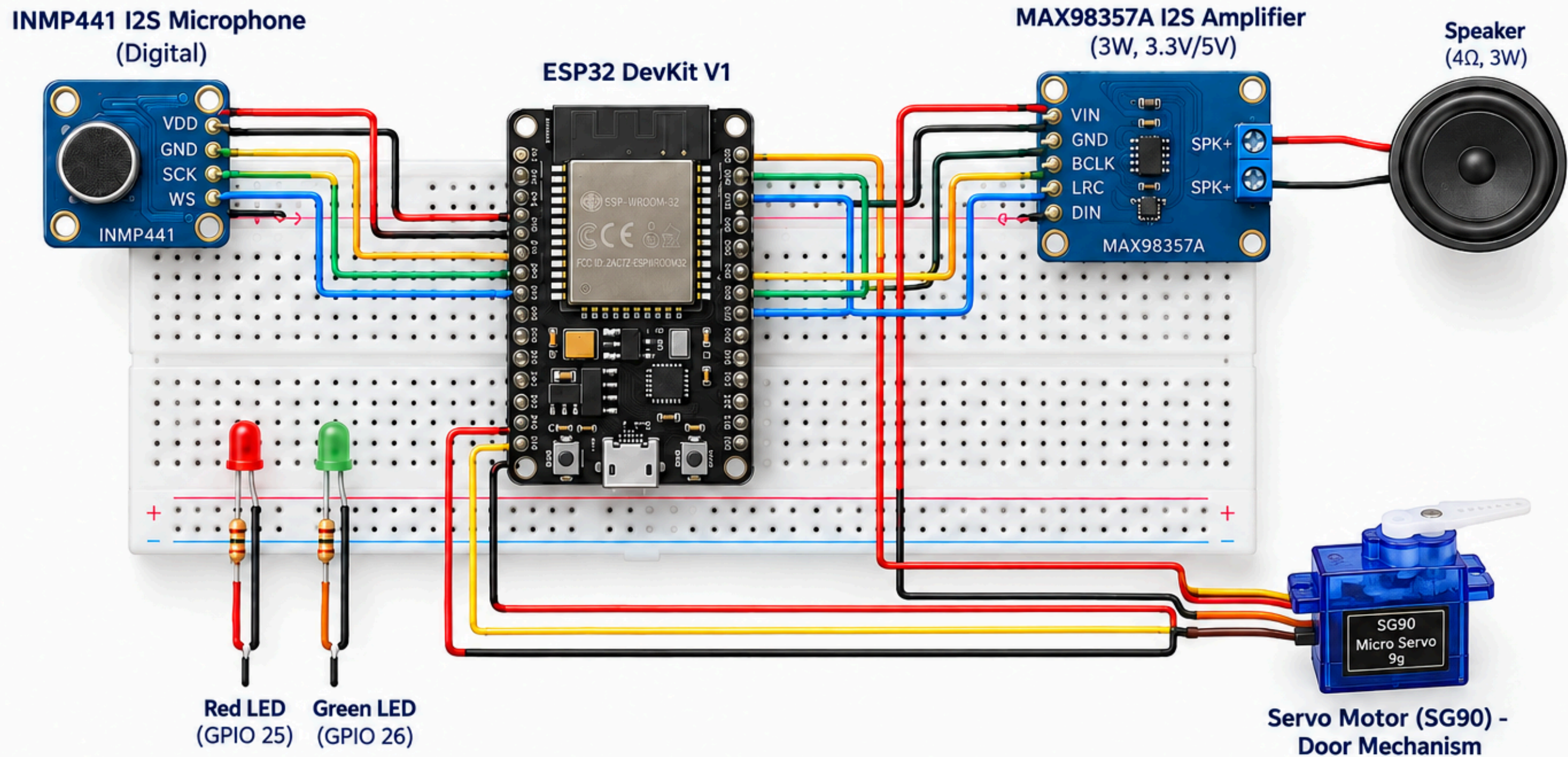
2. Key Features

- Voice AI: Natural Speech-to-Speech interaction.
 - Live Validation: Instant booking check via secure backend database.
 - Automated IoT Access: ESP32-powered door/turnstile unlocking.
 - Touchless: Safe, zero-contact patient experience.
- 
- 

SYSTEM DESIGN



HARDWARE & IOT



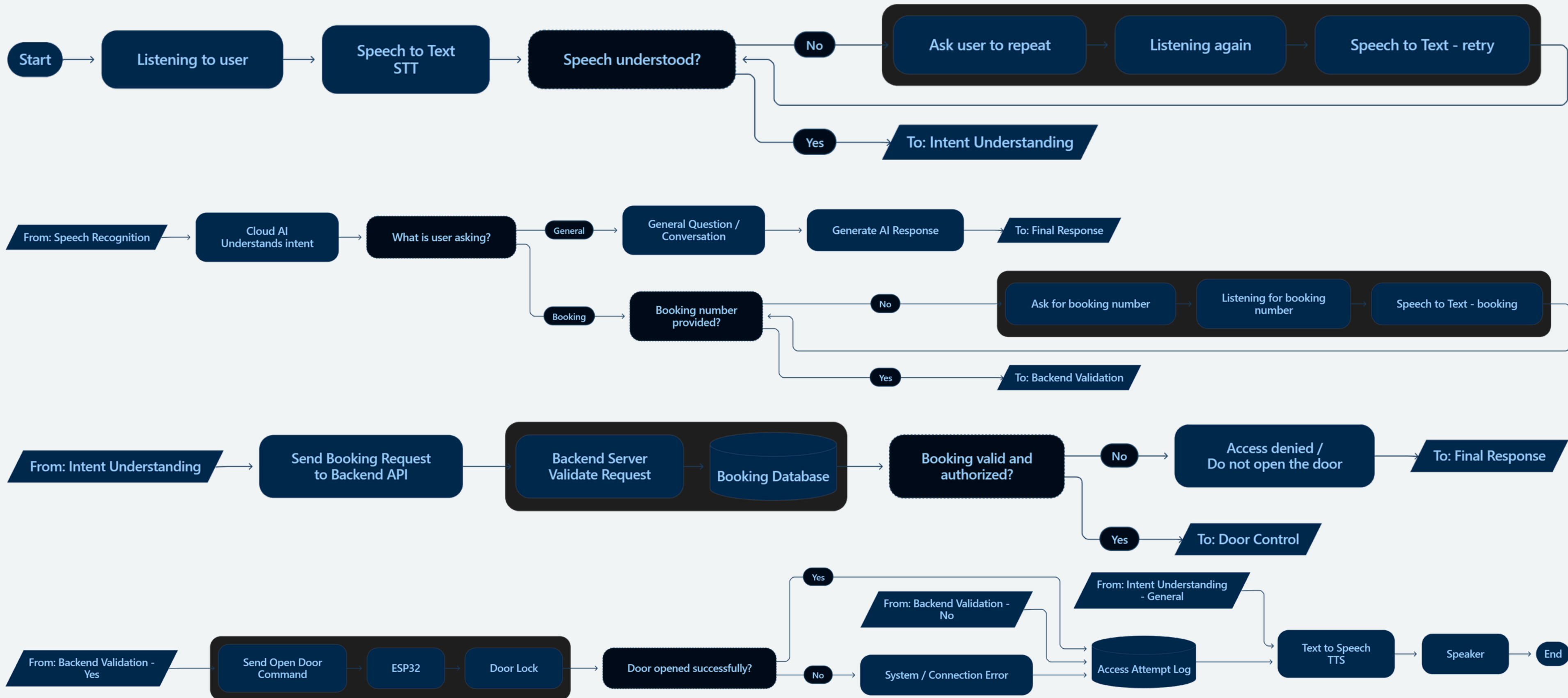
—	5V (VCC)
—	GND
—	I2S SCK (GPIO 14)
—	I2S WS (GPIO 15)
—	I2S SD (GPIO 32)
—	Servo Signal (GPIO 23)
—	LEDs (GPIO 25, 26)

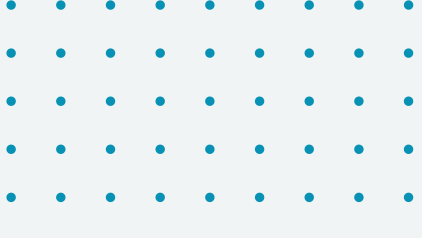
ESP32 Pin Usage	
GPIO 14	- I2S SCK (INMP441 & MAX98357A)
GPIO 15	- I2S WS (INMP441 & MAX98357A)
GPIO 32	- I2S SD (INMP441)
GPIO 23	- Servo Signal
GPIO 25	- Red LED
GPIO 26	- Green LED
5V / GND	- Power (all modules)

Notes:

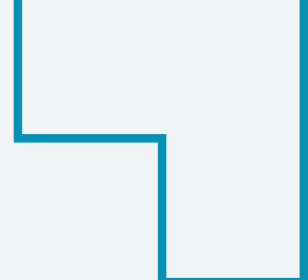
- SG90 Servo operates at 5V and is controlled via PWM signal on GPIO 23.

PROGRAMMATIC LOGIC





AI & DATA USAGE PLAN



1. Data Privacy & Handling

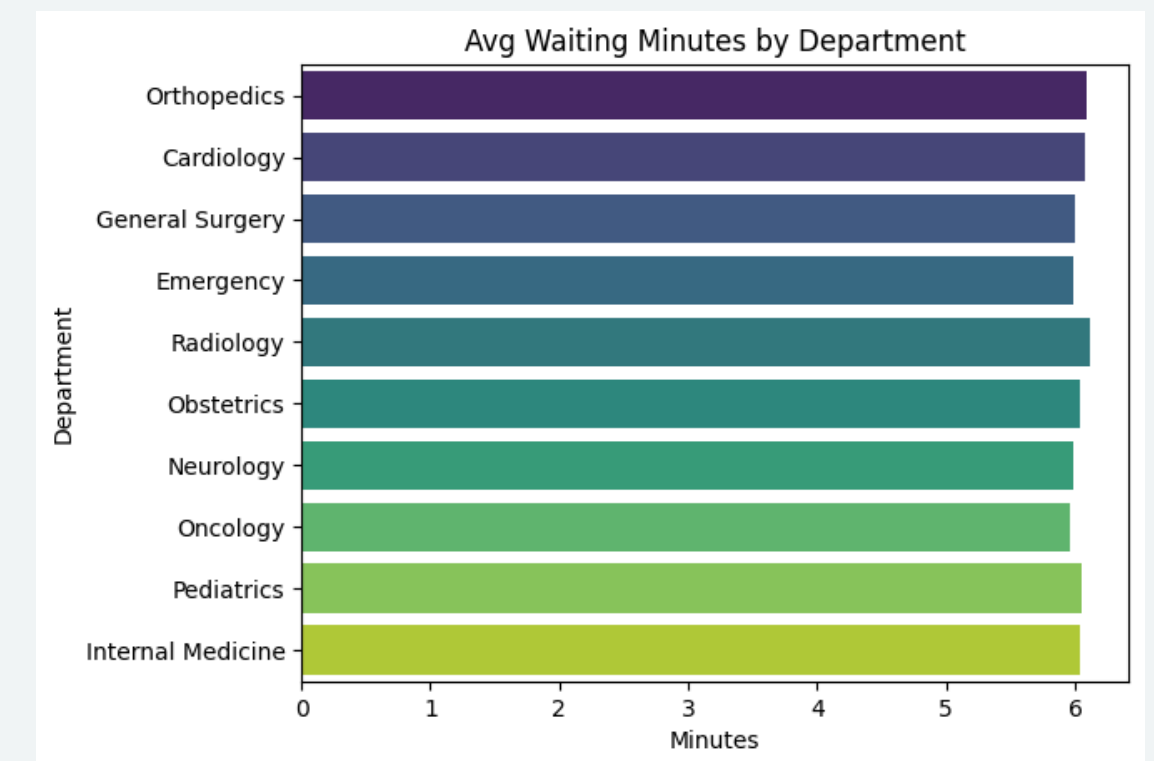
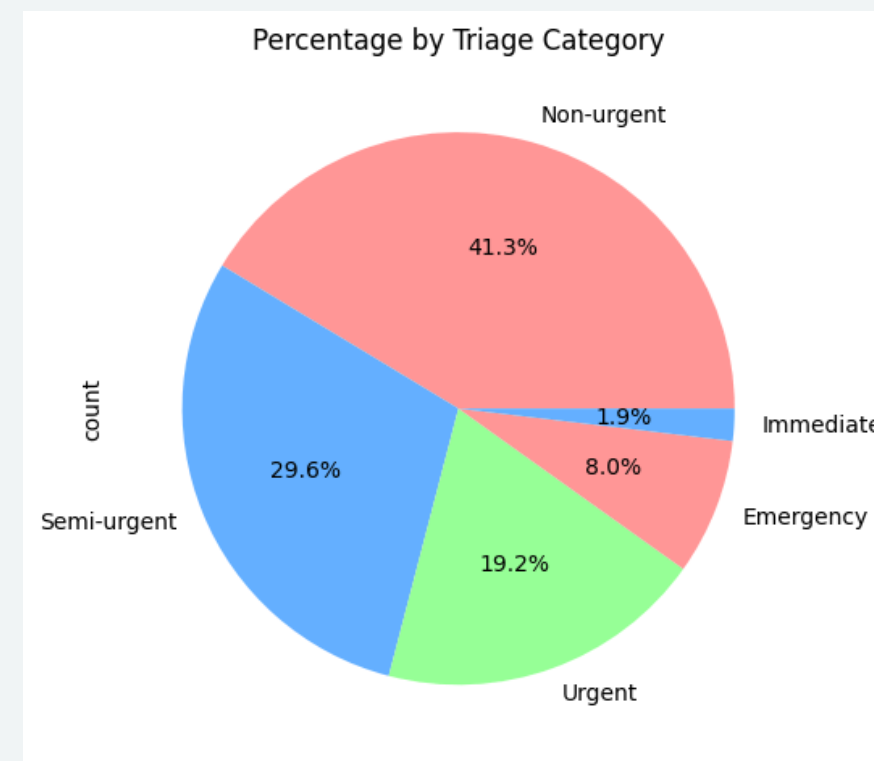
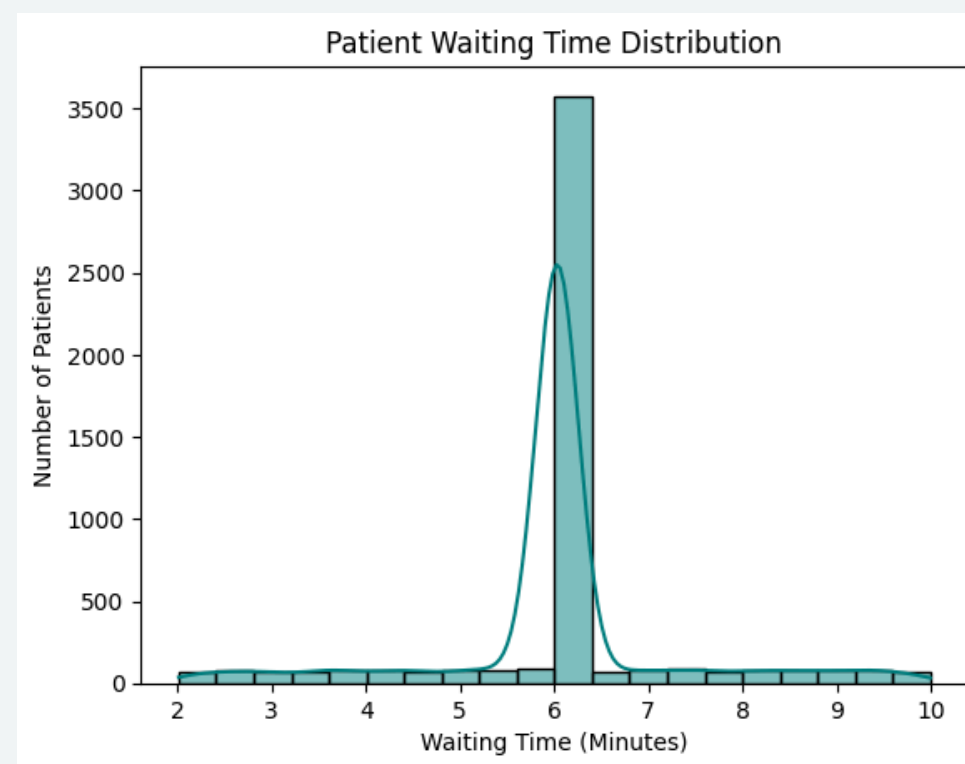
- Zero Audio Retention: Audio is processed (STT) in-memory and discarded immediately; no raw voice is saved.
- Structured Extraction: AI isolates only necessary entities (intent, booking_id) into strict JSON payloads.

2. AI Decision Engine (Tool Calling)

- Deterministic Control: The LLM uses standard Function Calling rather than free-form text to trigger actions:
 - check_booking(booking_id): Queries the database to verify the appointment.
 - open_door(): Sends an authorized signal to the ESP32 only upon database confirmation.
 - Anti-Hallucination: Physical access is strictly gated behind verified database states, completely eliminating unauthorized openings.
- 

DATA-DRIVEN RESEARCH & PREDICTIVE AI

- Dataset Analysis: Analyzed 5,000 hospital records, validating registration wait times as the core clinic bottleneck.
- Predictive ML Model: Trained a RandomForest AI model to dynamically calculate expected patient wait times.
- ROMI Integration: ROMI now informs the patient of their exact estimated wait time upon successful check-in.



Live Code Evidence: [On Google Colab](#)

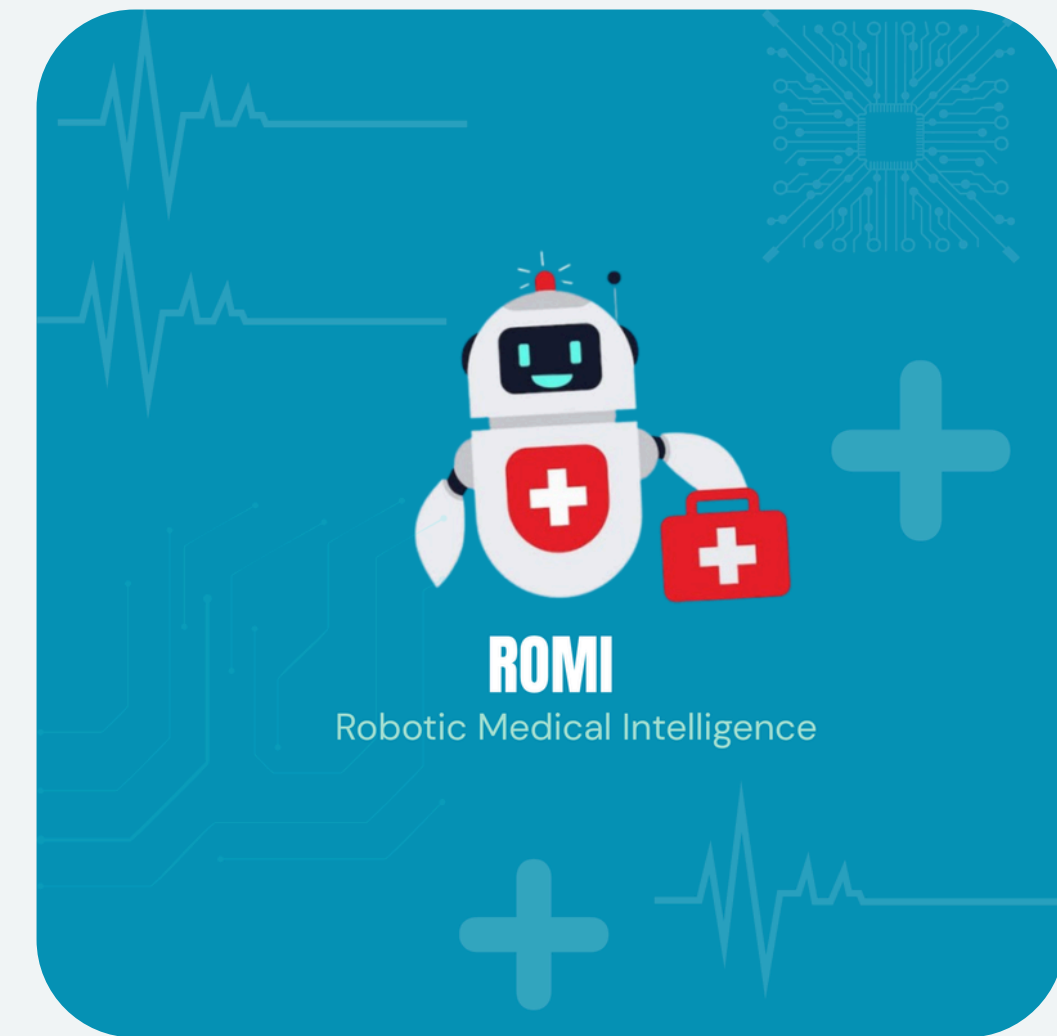
DIGITAL IDENTITY

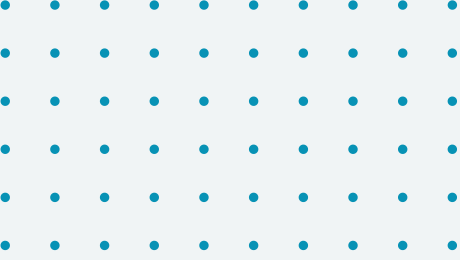
Name: **ROMI** (Robotic Medical Intelligence).

Identity: Friendly, reliable, clinical-grade AI assistant.

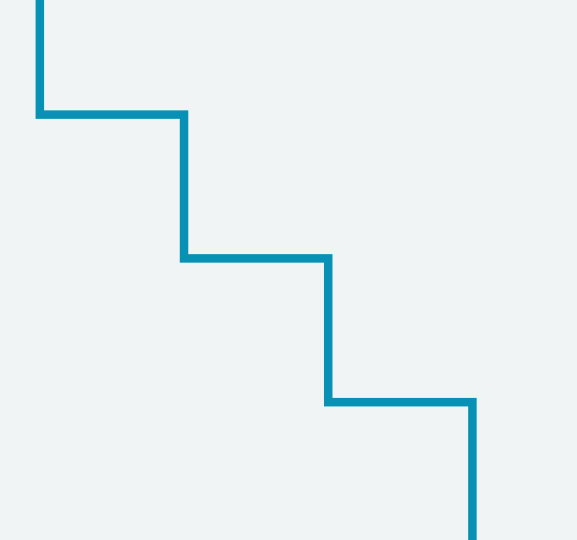
Colors:

- Teal & Mint: Tech reliability and patient comfort.
- Medical Red: Healthcare identity and urgency.
- Pure White: Hygiene and accessibility.





TRACK INTEGRATION

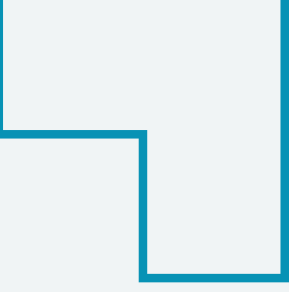
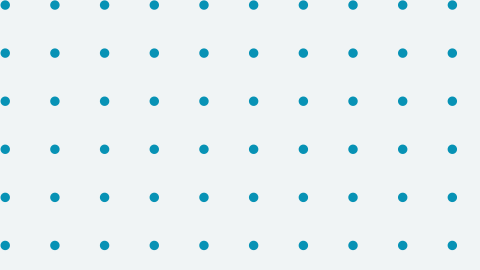


- Digital Arts (UI/UX): Designed the complete ROMI Digital Identity (Logo & Colors) and prototyped the dashboard interfaces on Figma.
- Web Development: Engineered the System Architecture, mapped the Programmatic Logic, and translated Figma designs into a deployed Next.js web application.
- Data Science: Conducted core medical-tech research, analyzed patient flow data, and developed the ROMI conceptual framework.
- Cybersecurity: Architected the ESP32 hardware circuits and implemented key threat mitigations:
 - AI Prompt Injection: Strictly separated logic; the AI cannot trigger hardware directly, only the backend verifies IDs.
 - Unauthorized IoT Access: The ESP32 rejects all network requests except those with our encrypted Webhook API Keys.
 - Data Exposure: Enforced Row Level Security (RLS) in the database to prevent mass extraction of patient records.



Live Demo: romi-demo.vercel.app





BUSINESS MODEL

1. Target Customers (B2B)

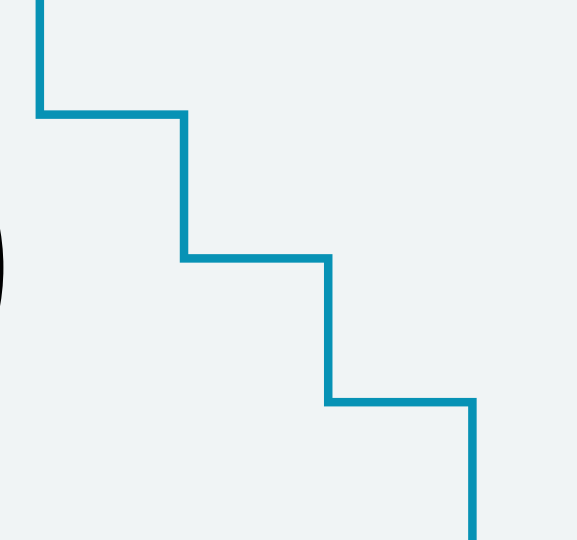
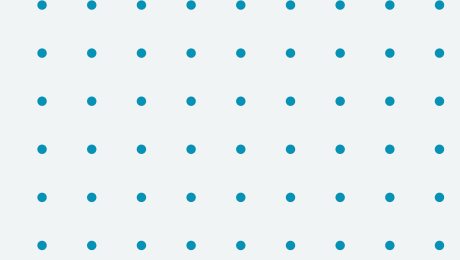
- Private Clinics & Hospital Outpatient Departments.

2. Value Proposition (ROI)

- Cost Reduction: Cuts receptionist workload by 60%.
- Efficiency: Eliminates waiting room queues & data errors.
- Brand Image: Tech-forward, premium patient experience.

3. Revenue Streams (Our Profit)

- Hardware Margin: One-time profit from initial kiosk setup.
 - SaaS Subscription: Monthly recurring revenue (high net profit after covering LLM/DB costs).
 - Maintenance: Annual fees for support & updates.
- 

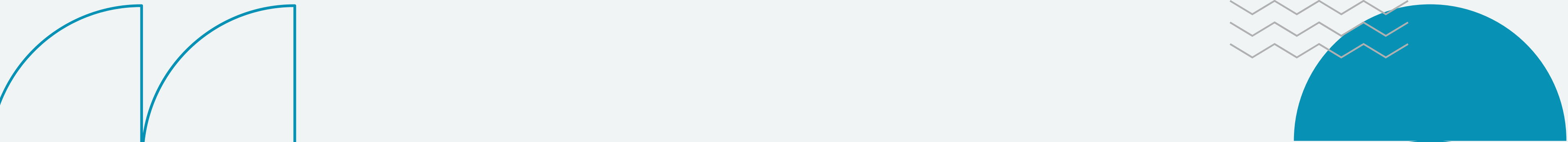


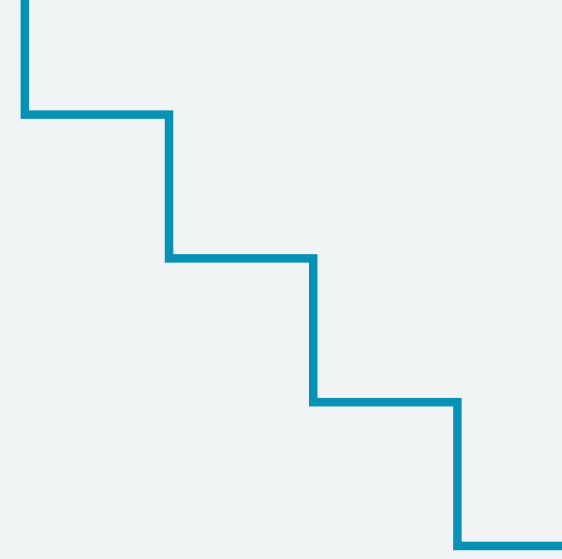
EXTRA POINTS (BONUS)

1. Advanced AI Feature: Multi-Turn Memory & Tool Calling

- Dynamic Flow: Handles unstructured Arabic dialogue, automatically asking follow-up questions if booking details are missing.
- Structured Output: Converts natural voice intents into precise JSON function calls for secure backend validation.

2. Advanced System Feature: Hardware Error Handling & Logging

- Robust Fallbacks: Detects ESP32 connection drops instantly, alerting the patient gracefully via TTS without system crashes.
 - Traceability: Every AI decision and physical door trigger is securely logged in real-time for clinical-grade security.
- 



THANK
YOU

